

SITE-SPECIFIC SEASONAL BASELINES FOR FIRE RISK IN ONTARIO

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Quantifying the spatiotemporal seasonal effects of forest fire regime characteristics is important for fire management planning. Of particular interest is the determination of when and where fires may occur; investigating whether fire risk has changed over time, and specifically determining regional seasonality effects. Records of forty-four years of data on fire occurrence in the province of Ontario's Intensive Fire Management Zone are used to develop additive spline smoothing-based models of forest fire ignition processes. The study data sets are recorded on an irregular grid that partitions the area into an irregular lattice whose elements are relatively homogeneous with respect to forest ecosystems, climate, and levels of fire suppression activities. Our methods provide simultaneously estimated predictions of site-specific seasonality components and an overall interannual trend component, as well as confidence intervals for these effects. The model is found to fit the data well. Spatial similarities in the seasonality trends are observed, and a visualization technique that explores spatial patterns in temporal trends, based on the singular value decomposition of the data matrix is applied.

La quantification des effets saisonniers spatiotemporels des caractéristiques d'un régime de feux de forêt est importante pour la planification de la gestion des feux. La détermination du moment et de l'endroit où un feu peut survenir; les recherches à savoir si les risques d'incendie ont changé à travers les années et la détermination des effets régionaux des cycles saisonniers sont d'un intérêt particulier. On utilise l'équivalent de 44 ans de données sur les feux qui se sont produits dans la zone intensive de feux de la province de l'Ontario dans le but de développer des modèles de processus de déclenchement de feux de forêt basés sur le lissage par spline. Les jeux de données à l'étude se trouvent dans un tableau irrégulier qui segmente l'aire en un treillis irrégulier dont les éléments sont relativement fidèles aux écosystèmes, au climat et au niveau d'activité de suppression d'incendie des forêts. Nos méthodes nous fournissent des prédictions des composantes saisonnières propres à l'endroit estimées simultanément ainsi que des composantes des tendances générales interannuelles; elles nous fournissent aussi des intervalles de confiance pour ces effets. Le modèle correspond bien aux données. Des similarités spatiales dans les tendances saisonnières sont observées et on applique une technique de visualisation qui explore les modèles spatiaux selon les tendances temporelles basées sur la décomposition en valeurs singulières de la matrice de données.

1. Introduction

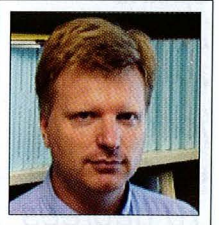
Improvements in fire management resource allocation decisions, budgeting, and forecasting depend on proper identification of spatiotemporal characteristics of forest fire seasonality. Such seasonal effects are influenced by several factors, but climate and vegetation are clearly very important factors. Current and recently observed weather conditions directly affect the flammability of a forested area. Moreover, climate indirectly affects a local fire regime by influencing the type of vegetation that covers a region. Regional vegetation is also influenced by land use activities such as forest harvesting and regeneration. The composition of vegetation in a region influences fire management because both living and dead forest vegetation are

potential fuels that play an important role in determining the behavior of any fires that ignite and burn in the area.

Locally, weather is a highly variable process on both spatial and temporal scales. However, it also follows an interannual pattern that induces a seasonality component on the numbers and sizes of fires in a region. For example, the climate in the late fall through to early spring in Ontario is not conducive to the ignition and spread of wildland fire and fire ignition processes "switch off" during this period. Conversely, from early spring to mid-fall of each year, Ontario experiences what is referred to as its *fire season* during which fire is present and active throughout its forested landscape. During this



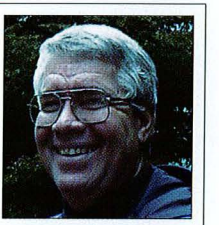
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...the
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period the common sources of forest fire ignitions are cloud-to-ground lightning and human activity. The interaction between these temporal and spatial effects produces a temporally dynamic fire regime which is not spatially homogeneous across the province of Ontario.

Temporal models for predicting human-caused fire occurrence in Northern Ontario were developed by *Cunningham and Martell* [1973], *Martell et al.* [1987], and *Martell and Belivacqua* [1989]. The results presented in the first paper indicated it is reasonable to model daily people-caused fire occurrence as a Poisson process with the average number of fires per day a decreasing function of fine fuel moisture. The second paper constructed logistic models for ignition risk, focusing on three distinct "subseasons" of the fire season. The latter paper extended those models by explicitly incorporating seasonal variability through trigonometric polynomial functions of day-of-year. Recent work [Brillinger et al. 2003; 2004; 2006; Preisler et al. 2004] constructed logistic generalized additive models (GAMs) on a fine space-time scale to approximate the spatiotemporal intensity of the corresponding point-process conditional on observed covariates, which are incorporated as smooth or linear additive effects.

In the literature discussed above, a single seasonality component is estimated but the Intensive Fire Management Zone of the province of Ontario (our study area) spans almost 25 degrees of longitude and almost 10 degrees of latitude. It covers a number of ecoclimatic zones, and the time of snowmelt which heralds the start of the fire season and the first snowfall which ends it, vary across the fire region. We therefore address the need to allow our seasonality component to vary spatially and we do so by simultaneously estimating several site-specific seasonality effects for a defined set of fire management compartments that partition our study region along with a global interannual trend. Hence, we can illustrate spatial and temporal similarities and dissimilarities in fire ignition rates across a vast landscape, while investigating for overall annual trends across the entire region. We also obtain standard errors for these estimates. Our analysis of Ontario fire data provides an illustrative example. We consider a large area that consists of the nearly 400 000 square kilometres of forested land that comprises Ontario's intensive fire management zone with forty-four years of fire records.

In this paper, we present a set of quantitative methods that could be employed in a geomatics toolkit for investigating spatial patterns in temporal trends. The statistical methods and models we describe could be used by planners in forest fire

management agencies to visualize trends in geographically referenced data. Our interest lies in the visualization of the spatially varying seasonality component in forest fire risk. In particular, we model temporal trends in the number of "fire days" using a generalized additive model. Site specific interannual seasonality terms are simultaneously estimated along with a global intra-annual trend term. Section 2 discusses the motivating context, with a description of the study area and some features of the data. The generalized additive model is summarized in Section 3. Its application to the Ontario forest fire data follows in Section 4, where variations in the rates are characterized by maps displaying site-specific seasonal components. Spatial similarities in some of these seasonality trends are observed. Section 5 discusses and applies a possible technique to explore these spatial patterns in temporal trends based on the singular value decomposition of the data matrix. We then conclude with a discussion of the results and of potential future work.

2. The Data and Study Area

The Aviation and Forest Fire Management Branch (AFFMB) of the Ontario Ministry of Natural Resources (OMNR) coordinates forest fire management activities on crown (i.e. government-owned) land and maintains detailed records of each reported fire. Historically, their management strategy was based on a partition of Ontario's fire region into three different large fire management zones, each of which received different levels of protection according to specified mandates. However, regardless of any prespecified level of protection, forest fire occurrences are spatially and temporally heterogeneous during and between fire seasons. Possible factors leading to these differences include regional differences in vegetation, weather, lightning, and human land-use patterns.

To account for this spatial heterogeneity, R. Kruus, D.L. Martell and B.M. Wotton of the Fire Management Systems Laboratory in the Faculty of Forestry at the University of Toronto constructed a map that partitions Ontario's Fire Region into a set of internally homogeneous polygons. That map was formed by overlaying the OMNR's Fire Management Zone coverage with the *Ecological Stratification Working Group's* [1996] ecoregions. That partition was further subdivided to produce a set of polygons that were roughly the same size, denoted as *Fire Management Compartments* (FMCs). We use this set of FMCs as the spatial units for our analysis and assume that each polygon

is relatively homogeneous in terms of its climate, vegetation and level of fire protection.

Finally, although it was not a significant policy change, we note that a new fire management strategy was recently introduced [OMNR 2004]. Consequently, to avoid any potential sources of bias, we have restricted our analysis to a period where the fire management policy was consistent, namely 1960 through 2003. Moreover, since the focus of our study is on areas with active management, we consider only FMCs in the area where all fires are suppressed: the Intensive Fire Management Zone.

Our data consist of weekly counts of the number of fire days for each of the sixteen FMCs in Ontario's Intensive Fire Management Zone during the fire seasons from 1960 through 2003. A fire day is as a day during which one or more forest fire ignitions are reported in the specified FMC. The ignition location of each fire is based on digital records of Ontario forest fire data provided by the OMNR. The weekly counts of fire days for each year in each of the FMCs under study were aggregated in R [R Development Core Team 2008]. Ontario's managed forest region, the FMC partition, and the Intensive zone are illustrated in Figure 1.

3. Generalized Additive Models and Penalized Spline Smoothers

Generalized additive models (GAMs) [Hastie and Tibshirani 1990] extend the popular generalized linear models (GLMs) [McCullagh and Nelder 1989] by allowing for nonlinear covariate effects through the incorporation of additive non-parametric smooth functions. Consider the univariate case with n observations indexed by i , $i = 1, \dots, n$. Assume the distribution of the i th response variable Y_i belongs to the exponential family and that its mean $\mu_i \equiv E[Y_i]$ is related to explanatory variables $x_{1i}, x_{2i}, \dots$ via a "link function" $h(\cdot)$. Letting $\eta_i \equiv h(\mu_i)$, a GAM has the following general structure

$$\eta_i = \mathbf{X}_i \boldsymbol{\beta} + \sum_{m=1}^M s_m(x_{mi}), \quad (1)$$

where $s_m(\cdot)$ are smooth functions (commonly referred to as "partial effects" or "smoothers") of the covariates x_{mi} ; $\mathbf{X}_i$ is the i th row of the design matrix for any fixed linear effects, whose p coefficients are contained in the vector $\boldsymbol{\beta}$, which usually includes an intercept term.

In our work, the smoothers are modelled using linear expansions of basis spline functions. For example, a univariate smooth function $s(x)$ can be represented as

$$s(x) = \sum_{k=1}^K \phi_k b_k(x),$$

where ϕ_k are unknown constants which are estimated from the data, and the $b_k(x)$ are piecewise polynomials, joined at a pre-specified set of K "knots." These knots form a partition across the range of the covariate in question. In other words, $s(x)$ can be represented as a linear combination of known functions, and hence, estimating s is equivalent to estimating the coefficients ϕ_k . This linear representation facilitates estimation via the likelihood-based inference used for GLMs.

Choosing an appropriate knot sequence is not trivial. An alternative approach is to use a relatively large number of knots (often located at the observed x -values) and to incorporate penalty components in the likelihood to control the amount of smoothing. Given the additive characteristic of the basis expansion, one commonly penalizes the total amount of roughness in s with a penalty term, such as

$$\lambda \int [s''(x)]^2 dx,$$

where $s''(x)$ denotes the second derivative of the function, and the parameter λ is to be chosen using an appropriate goodness of fit criterion. Given λ , the model-fitting objective function becomes

$$\sum_{i=1}^n \left[y_i - \mathbf{X}_i \boldsymbol{\beta} - \sum_{m=1}^M s_m(x_{mi}) \right]^2 + \sum_{m=1}^M \lambda_m \int [s_m''(x)]^2 dx_m.$$

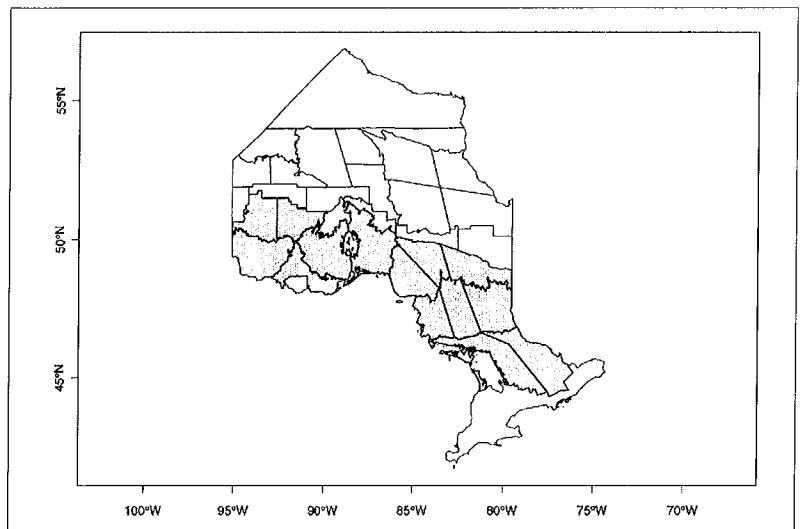


Figure 1: The Fire Management Compartment (FMC) partition of Ontario's fire region, and the intensive fire management zone (shaded).

This is to be minimized with respect to the fixed linear parameters β and the set of basis coefficients $\{\phi_k\}_{k=1,\dots,K}$. The penalty terms $\{\lambda_m\}_{m=1,\dots,M}$, are chosen so that the Generalized Cross Validation (GCV) score [Craven and Wahba 1979; Golub et al. 1979] is minimized. This GCV score is essentially a numerically efficient modification of the ordinary cross validation score, where the average squared prediction error is estimated over all data sets where a single observation is omitted when fitting the model, and then the observation is predicted. Technical details on modelling and GCV estimation for GAMs with multiple smoothing parameters appears in Wood [2000; 2004; 2006].

Given indices for time and space, nonlinear spatiotemporal seasonality effects can be incorporated into models in the form of (1) in several ways. Two obvious possibilities include modelling space and time together via a single multivariate smoother, or including space and time as separate additive smooth effects. Given areal data, an attractive alternative is to fit separate seasonality partial effect curves for each polygon. This is the approach we take in our analysis.

4. Application to Forest Fire Day Counts

We fit the following model to weekly counts of fire days for the FMCs in Ontario's Intensive Fire Management Zone for the years 1960-2004:

$$\text{logit}(p_i) \equiv \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum_l f_l(w_i) I_{\{FMC_i=l\}} + s(y_i),$$

where p_i represents the probability of a fire day during week w_i of year y_i in FMC_i , $i = 1, \dots, n$, indexing the list of observations; β_0 is an overall intercept term in the model; $I_{\{FMC_i=l\}}$ is an indicator (i.e. a

dummy variable) that the i th observation was in FMC l ; $f_l(w)$ are FMC-specific smoothers of week; and, $s(y)$ is a smooth intra-annual trend effect for the entire study area. The site-specific smoothers were estimated using cyclic cubic regression splines so that the start and end of each interannual seasonality component would match. The annual trend smoother was estimated using a cubic regression spline. All smoothers in the model contained 10 equally spaced knots. The model was estimated in R [R Development Core Team 2008], using the gam function in the mgcv package [Wood 2006].

The estimated intercept is -4.85, has a standard error of 0.08, and is highly significant. Its sign and magnitude reflect the fact that forest fire ignitions are rare events over the course of each year. The estimated interannual trend term is plotted in Figure 2. Although statistically significant, this trend term has little impact on the fire risk probabilities as reflected by its range of values, relative to the intercept. These estimates are given on the scale of the linear predictor (i.e. the logit scale).

Figure 3 presents the estimated site-specific baseline probability of a fire day for each of the FMC polygons. Approximate 95% pointwise confidence regions are also given for each polygon. All presented curves are statistically significant at the 5% level.

A seasonal pattern is evident. In some cases the fire risk is bimodal, while in others, primarily more northerly and hence more remote, the fire risk appears to be unimodal. This can be explained by human land use and weather patterns in Ontario. For example, each year Ontario experiences a short period of increased fire danger before new leaves (with high moisture content) appear on the deciduous trees and the herbaceous vegetation and shrubs "green up." During this time, there is increased solar radiation and wind penetration to the forest floor, which can dry the fine litter fuels, producing an environment that is more conducive to both fire ignition and spread. Lightning is not that common during this period and hence, does not account for many of these early season fires. Instead, it is human activity, such as rural residents and recreational campers for example, which cause most ignitions during this period. A possible explanation for the unimodality in the more northerly FMCs could be decreased human activity in these more remote regions.

After the appearance of leaves, the temperature remains relatively cool and hence fire risk is reduced because of decreased drying of the forest floor. As the summer progresses, the temperature tends to rise and periods of drought lead to substantial fire risk once again. Moreover, Ontario's

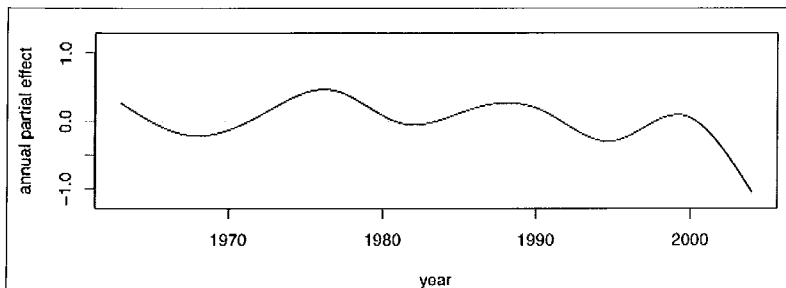


Figure 2: Estimated annual partial effect for the entire study region, plotted on the scale of the linear predictor (i.e. logit scale). Shading shows the 95% confidence region.

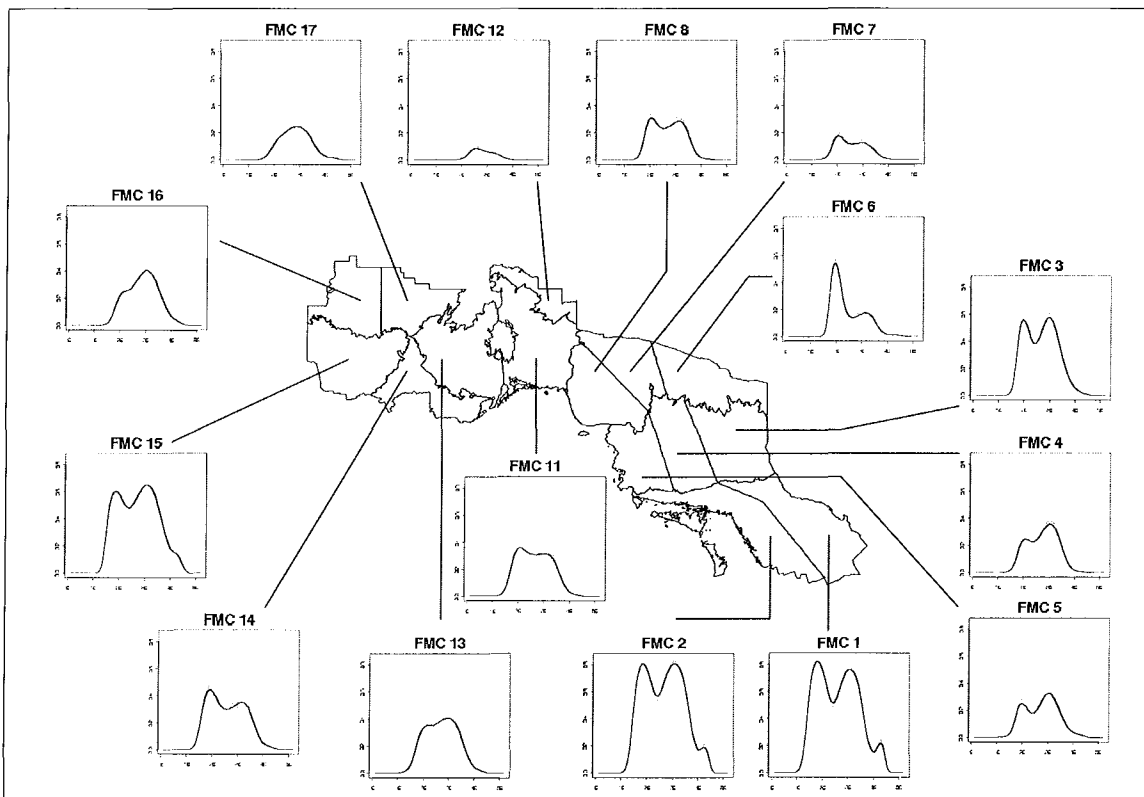


Figure 3: Estimated probability of a fire day in a given week for each Fire Management Compartment in Ontario's Intensive Fire Management Zone. Shading shows the 95% confidence regions.

weather pattern during the peak of the summer includes a high frequency of electrical storms. The combination of human-caused and lightning-caused ignition rates is likely causing the bimodality observed in the less remote FMCs.

To assess the fit of our model, we compare observed and expected counts of fire days aggregated at different scales. Here, the expected counts are obtained by summing the fitted values that have been transformed to probabilities using the inverse of the logit link. Plots of aggregated observed and expected counts of fire days appear in Figure 4, which provides results when counts are aggregated weekly for each of the FMCs in our study, and for when counts are aggregated annually. In these plots the observed number of fires appear as points, while those expected under the models appear as lines. The model appears to fit the data well for all locations. Similarly, the fit appears reasonable when counts are aggregated and viewed across years.

Our analysis reveals interesting trends in the spatial context; in several cases, neighbouring curves are somewhat similar. Recall that the FMC partition was based on both ecoregions and fire management strategies, so some neighbouring FMCs span the same ecoregion. Consequently, these similarities are not unexpected. For example, FMCs 1 and 2 (the most southeasterly FMCs) are in the same ecoregion

and have very similar shapes. We note, however, that techniques for the clustering of curves accounting for spatial contiguity or another neighbourhood structure are not readily available. In the next section we investigate spatial patterns in temporal trends using a recently proposed and relatively simple exploratory technique.

5. A Singular Value Decomposition Method

Recently, Guttorp suggested the use of a geomatics tool for exploring spatio-temporal trends in spatial data. This technique is based on the singular value decomposition (SVD) of the data matrix. Its objective is to distill a set of independent curves, each of which is a component of the overall temporal trend. The temporal behaviour at each site can then be modelled in terms of transformations based on these nonparametric curves.

Let Z denote a $n \times m$ matrix of data, where each column consists of the time series of response values at a site. The singular value decomposition is given by

$$Z_{(n \times m)} = U_{(n \times m)} D_{(n \times m)} V_{(m \times m)},$$

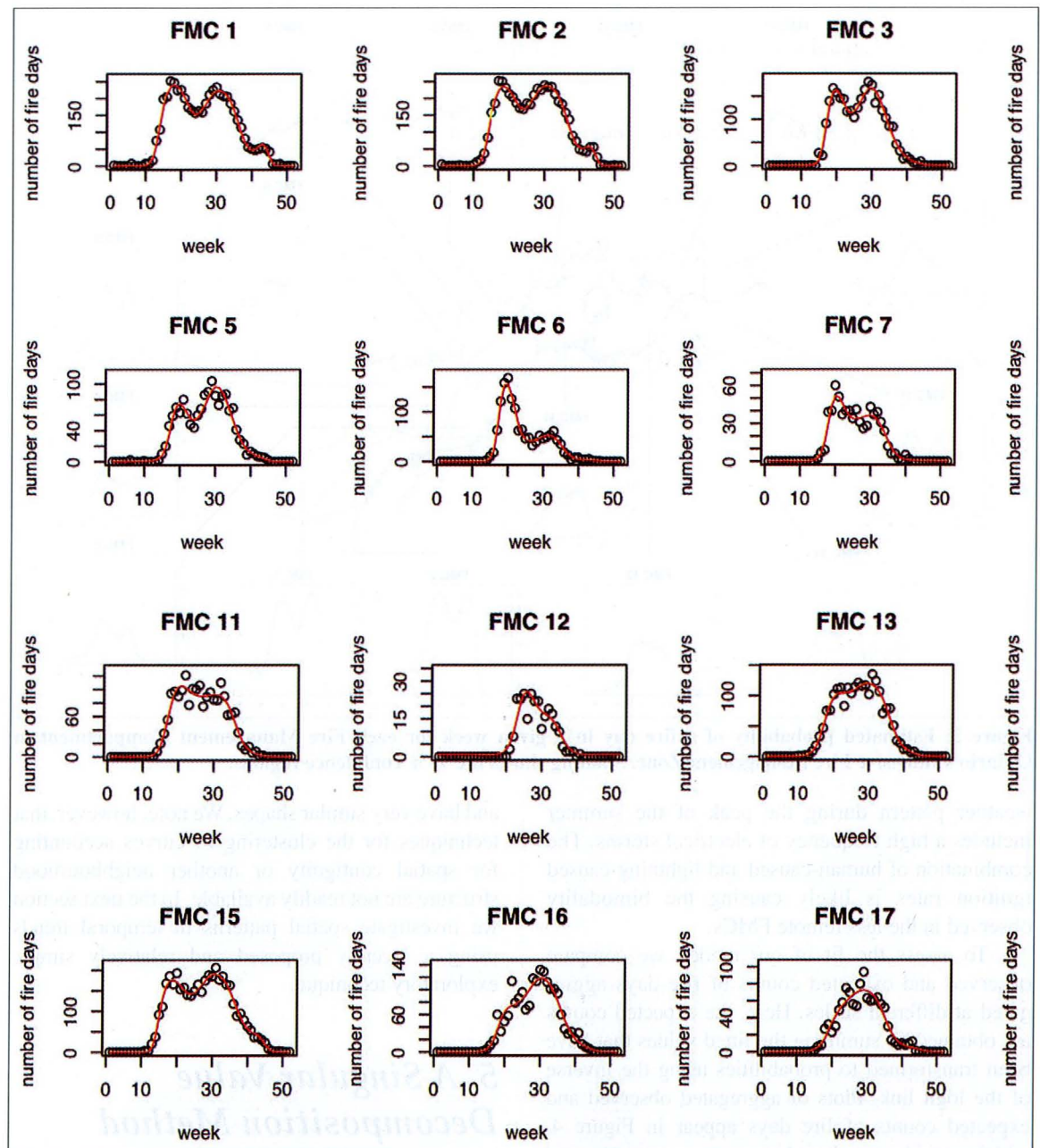


Figure 4: Comparison of the total observed number of fires (points) versus those predicted by the model (solid line) for each term in the model. In the plots for each of the FMCs, the counts are aggregated weekly. In the plot for the annual trend term, the counts are aggregated yearly.

where the bracketed subscripts in the above denote the number of rows and columns in each respective matrix. The matrices U and V have orthogonal columns and D has nonzero entries on the main diagonal only (the *singular values*).

The columns of U that correspond to the largest singular values are smoothed to produce a set of temporal basis functions, which are used to transform time for use in regressions of site-specific responses against time. This is a fully nonparametric alternative to spline regression, but can be thought of as analogous to it. Moreover, this methodology

is similar to principal components analysis (PCA) and the regression of responses on principal components, both of which are proven useful exploratory techniques in data analysis [e.g. Maindonald and Braun 2007, ch. 12-13].

As in PCA, the singular values provide a measure of the amount of variation in the data “explained” by a given singular vector. Guttorm proposed this SVD method for modelling space-time means, and in particular, as a technique to explore for spatial patterns. The latter is achieved by producing choropleth maps based on the estimated

site specific coefficients from the regressions on the first few SVD temporal basis functions.

In our context, Z is a matrix whose columns are the time series of the number of fire days per week for each of the FMCs in Ontario's Intensive Fire Management Zone. Here, we present results using the first four SVD temporal basis functions which are illustrated in Figure 5. We note that the first temporal basis function (panel (a)) is clearly capturing an intra-annual seasonality, with larger amplitudes in the mid 1970s and 1990s. The other basis functions (panels (b)–(d)) modulate the amplitudes in various years, and are somewhat suggestive of longer-term cycles. In particular, the positive peaks in the third basis function (panel c) reflect the shape in the smoother for year displayed in Figure 2.

Similar to the idea of regression of responses on principal components, we fit over dispersed Poisson GLMs [McCullagh and Nelder 1989] for each of the FMCs in our study. The FMC-specific time series of weekly counts of fire days were regressed against the first four SVD temporal basis functions. We use the first four basis functions for illustration because together they explain nearly 50% of the variation in the data. We remark, however, that the first three basis functions account for about 47% of the variation, and focus our discussion mainly on these. (As an aside, notice that the fourth basis function in Figure 5, panel (d) does not exhibit as large as amplitudes in its peaks; hence, it doesn't modulate temporal behaviour as much as the other basis functions illustrated in that figure.) This yields sets of spatially referenced coefficient estimates, which we explore in Figure 6. Each panel in this Figure 6 is a choropleth map for each of the four coefficient estimates. The gradient in the shading of each of the plots represents the size of the respective coefficient, with lighter being smaller and darker being larger. The actual values of the coefficients are not required, since in this context, the relative values reflected by the colour-coding provide a means for visualizing the data to look for spatial patterns.

Panel (a) of Figure 6 corresponds to the coefficient of the first temporal basis function, with its clear seasonal cycle. This map indicates that the FMC whose seasonal component is dominated by this basis function is in the northwest of the study area, while three FMCs in the northeast do not have as strong a connection to this type of temporal trend. Panel (b) shows that the eastern region of the study area is more tempered by the second temporal basis function than the west; an east-to-west gradient is clear in this case. The remaining two panels in the figure do not exhibit strong spatial patterns.

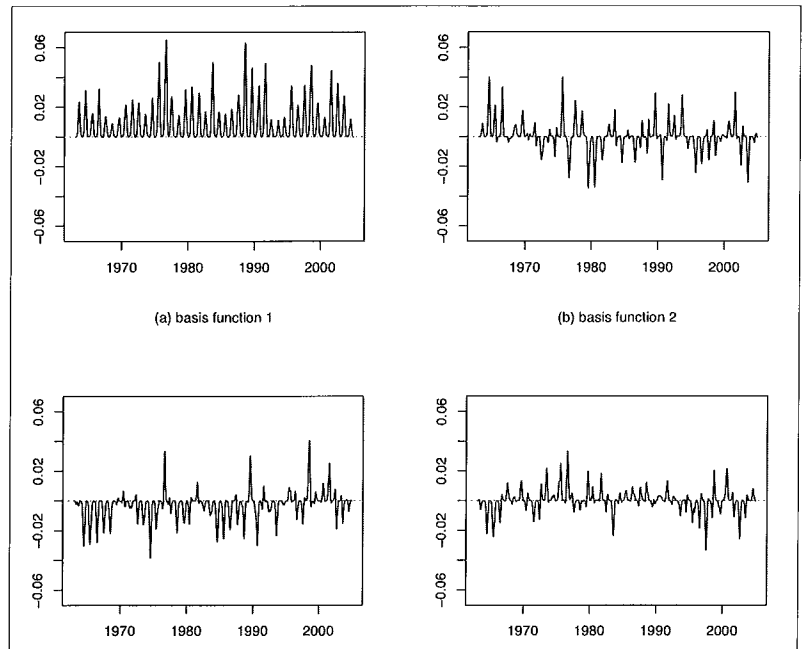


Figure 5: Plots of the first four temporal basis functions estimated using the singular value decomposition (SVD) methodology.

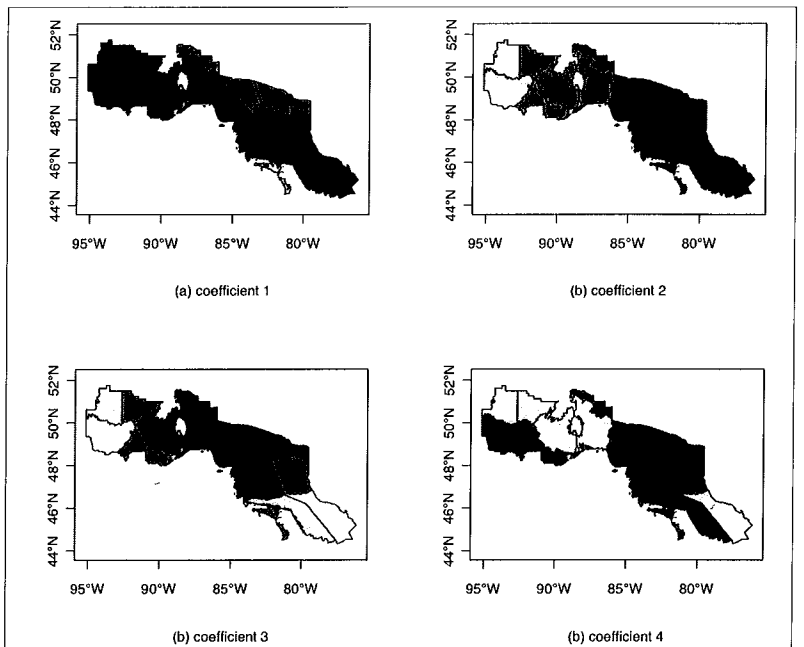


Figure 6: Choropleth maps based on the estimated coefficients obtained when the fire counts for each FMC polygon are regressed on the first four SVD-based temporal basis functions. The gradient in the shading of each of the plots represents the size of the respective coefficient, with lighter being smaller and darker being larger.

6. Discussion

In this article we have demonstrated how statistical modelling can be used to underlie geomatics tools for investigating spatial-temporal effects

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by presenting smoothing methods and models for spatially referenced forest fire ignition data. The development of such tools may be useful for fire management agencies, where the visualization of such patterns could aid resource allocation decisions. In particular, we have shown how binomial generalized additive models can be used to estimate site-specific non-linear seasonality trends in fire ignition risk. Spatial heterogeneities in our large study region were accounted for by incorporating site-specific temporal partial effects, where the units of our spatial partition are such that they can be assumed to be internally homogeneous with respect to their climate, vegetation, and level of fire protection. Geographical variations in this seasonality component were characterized using maps which display differing trends. We found that our model fitted the data well when we compared observed counts of fire days, versus those expected under the fitted model. This extends the past models in the literature, which incorporated a single seasonality effect when studying smaller regions.

The spline-based smoothers employed in our generalized additive model provide a more flexible approach than those using periodic functions, which have appeared in the historical literature when modelling seasonal trends. We also note that, besides seasonal trends within each fire season, trends across years are of also of interest to the fire science community. For example, *Brillinger et al.* [2006] examined the incorporation of a random effect component to account for interannual variability and suggested that a smoother of year might be necessary. We included the latter in our model and found that the overall interannual trend component, although significant, was not suggestive of any consistent upwards or downwards trend in fire risk over the period of our study. (Note that although the estimated annual effect appears to decrease in recent years, no definite conclusion can be drawn, because the accuracy of spline smoothers decreases at the boundaries.)

Spatial similarities in seasonal trends were investigated further, using an exploratory technique based on the singular value decomposition of the data matrix. The result is a set of independent curves: the temporal basis functions. The temporal behaviour at each site can be explored by regressing the data for that site against these nonparametric curves and producing choropleth maps of the estimated coefficients at each site. The latter is a simple visualization tool which can be employed to explore spatial patterns in temporal trends. Further studies examining the temporal basis functions are necessary: the first basis function estimated in our study has a strong intra-annual seasonality compo-

nent, while the other functions are suggestive of longer-term cycles. Additional investigation is warranted before any definite conclusions could be stated. A more thorough understanding of the temporal basis functions would aid the interpretation of any spatial patterns observed in the choropleth maps that were presented.

Finally, we note that the hierarchical structure of the Ecological Stratification Working Group's ecological framework (1996) allows future work to be scaled to either the finer spatial scale of the ecodistricts, or to a coarser scale such as the eco-provinces or ecozones. Since this hierarchical structure also represents a national framework, one can also easily extend this type of modelling to any other region in Canada. It would be interesting to extend this model across the entire boreal.

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References

- Brillinger, D.R., H.K. Preisler, and J.W. Benoit. 2006. Probabilistic risk assessment for wildfires. *Environmetrics*, 17: 622–633. DOI: 10.1002/env.768.
- Brillinger, D.R., H.K. Preisler, and H.M. Naderi. 2004. Wildfire chances and probabilistic risk assessment. In *Proceedings of the Sixth International Symposium on Spatial Accuracy Assessment in Natural Resource and TIES*, Portland, Maine.
- Brillinger, D.R., H.K. Preisler, and J.W. Benoit. 2003. Risk assessment: a forest fire example. In *Science and Statistics: A Festschrift for Terry Speed*. (Ed. D.R. Goldstein). Institute of Mathematical Statistics Lecture Notes, 40: 177–196. Beechwood, OH.
- Craven, P. and G. Wahba. 1979. Smoothing noisy data with spline functions. *Numerische Mathematik* 31: 377–403.
- Cunningham, A.A. and D.L. Martell. 1973. A stochastic model for the occurrence of man-caused forest fires. *Canadian Journal of Forest Research*, 3(2): 282–287.
- Ecological Stratification Working Group. 1996. A National Ecological Framework for Canada. Agriculture and Agri-Food Canada, Research Branch, Centre for Land and Biological Resources Research and Environment Canada, State of Environment Directorate, Ottawa/Hull. 125pp. and Map at scale 1:7.5 million. <http://www.ec.gc.ca/soer-ree/English/framework.cfm>.

- Golub, G.H., M. Heath, and G. Wahba. 1979. Generalized cross validation as a method for choosing a good ridge parameter. *Technometrics* 21(2): 215–223.
- Guttorp, P. 2009. Lecture notes from the MITACS/GEOIDE Summer School on Mathematical and Statistical Descriptions of Forest Fire Spread: Spatial Statistics and Level Set Method. Held June 15–19, 2009 at the Hinton Training Centre in Alberta, Canada.
- Hastie, T. and R. Tibshirani. 1990. *Generalized additive models*. Chapman and Hall: London, U.K.
- Maindonald, J. and W.J. Braun. 2007. *Data Analysis and Graphics Using R: An Example-Based Approach* (2nd Ed.). Cambridge: New York, U.S.A.
- Martell, D.L. and E. Belivacqua. 1989. Modelling seasonal variation in daily people-caused forest fire occurrence. *Can. J. For. Res.*, 19(2): 1555–1563.
- Martell, D.L., S. Otukol, and B.J. Stocks. 1987. A logistic model for predicting daily people-caused forest fire occurrence in Ontario. *Can. J. For. Res.*, 17: 394–401.
- McCullagh, P. and J.A. Nelder. 1989. *Generalized linear models* (2nd ed.). Chapman and Hall: London, U.K.
- Ontario Ministry of Natural Resources (OMNR). 2004. *Forest Fire Management Strategy for Ontario*. Queen's Printer for Ontario, Toronto.
- Preisler, H.K., S. Chen, F. Fujioka, J.W. Benoit, A.L. Westerling. 2008. Wildland fire probabilities estimated from weather model-deduced monthly mean fire danger indices. *Int. J. Wildland Fire*, 17: 305–316.
- Preisler, H.K., D.R. Brillinger, R.E. Burgan, and J.W. Benoit. 2004. Probability based models for estimation of wildfire risk. *Int. J. Wildland Fire*, 13: 133–142.
- R Development Core Team. 2008. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL <http://www.R-project.org>.
- Wood, S.N. 2000. Modelling and smoothing parameter estimation with multiple quadratic penalties. *J. R. Statist. Soc. B*, 62: 413–428.
- Wood, S.N. 2004. Stable and efficient multiple smoothing parameter estimation for generalized additive models. *J. Amer. Statist. Assoc.* 99(467): 673–686. doi:10.1198/016214504000000980.
- Wood, S.N. 2006. *Generalized Additive Models: An Introduction with R*. Chapman and Hall, CRC Press: Boca Raton, FL.

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